

 Deployment Guide

SayOne HRIS Career Portal

Panduan Deploy ke Server Ubuntu

Frontend: Next.js 16 + TypeScript + Tailwind CSS

Backend: Spring Boot (Java 21) — terpisah

Repo: github.com/pujaasma0721/sayone-hris-portal

Versi: 1.0 | Tanggal: Juni 2026

1. Prerequisites

Sebelum memulai deployment, pastikan server Ubuntu Anda memenuhi requirements berikut:

Server Requirements

Komponen	Minimum	Recommended
OS	Ubuntu 22.04 LTS	Ubuntu 24.04 LTS
CPU	1 vCPU	2 vCPU
RAM	1 GB	2 GB
Disk	10 GB	20 GB SSD
Node.js	20.x	22.x LTS
Bun	1.1+	Latest
Nginx	1.18+	Latest

Yang Dibutuhkan

- **Akses SSH** ke server dengan user yang punya sudo privilege
- **Domain/subdomain** yang sudah pointing ke IP server (misal: `portal.sayone.my.id`)
- **Backend HRIS** sudah running dan accessible (misal: `https://hris.sayone.my.id`)
- **Portal slug** sudah di-generate dari admin panel HRIS (menu Career Portal → Generate Access Link)

INFO

Portal ini adalah **pure frontend** — tidak butuh database. Semua data di-fetch dari backend HRIS via API.

Update Sistem

```
# Login ke server via SSH
ssh user@your-server-ip

# Update sistem
sudo apt update && sudo apt upgrade -y
```

```
# Install toolsdasar  
sudo apt install -y curl wget git unzip build-essential
```

2. Install Node.js & Bun

2.1 Install Node.js 22 LTS

```
# Install Node.js via NodeSource
curl -fsSL https://deb.nodesource.com/setup_22.x | sudo -E bash -
sudo apt install -y nodejs

# Verifikasi
node --version    # harus v22.x.x
npm --version     # harus 10.x.x
```

2.2 Install Bun (Runtime)

```
# Install Bun
curl -fsSL https://bun.sh/install | bash

# Load bun ke PATH (tambahkan ke ~/.bashrc)
echo 'export BUN_INSTALL="$HOME/.bun"' >> ~/.bashrc
echo 'export PATH="$BUN_INSTALL/bin:$PATH"' >> ~/.bashrc
source ~/.bashrc

# Verifikasi
bun --version    # harus 1.1+
```

2.3 Install PM2 (Process Manager)

```
# Install PM2 global
sudo npm install -g pm2

# Verifikasi
pm2 --version
```

KENAPA PM2?

PM2 menjaga aplikasi Next.js tetap running (auto-restart jika crash, start on boot).

3. Clone & Build Project

3.1 Clone Repository

```
# Buat direktori aplikasi
sudo mkdir -p /var/www/sayone-portal
sudo chown $USER:$USER /var/www/sayone-portal

# Clone repo
cd /var/www/sayone-portal
git clone https://github.com/pujaasma0721/sayone-hris-portal.git .

# Verifikasi
ls -la # harus ada package.json, src/, next.config.ts
```

3.2 Install Dependencies

```
cd /var/www/sayone-portal

# Install dependencies dengan Bun
bun install

# Verifikasi tidak ada error
echo "Dependencies installed successfully"
```

3.3 Konfigurasi Environment

```
# Buat file .env.local
cat > /var/www/sayone-portal/.env.local << 'EOF'
# Backend API URL (HRIS Spring Boot)
NEXT_PUBLIC_API_URL=https://hris.sayone.my.id

# Default company code (fallback jika tidak ada cookie)
NEXT_PUBLIC_COMPANY_CODE=SAYONE

# Set false untuk pakai real backend (bukan mock)
NEXT_PUBLIC_API MOCK=false
EOF

# Verifikasi
cat .env.local
```

PENTING

Ganti `NEXT_PUBLIC_API_URL` dengan URL backend HRIS Anda. Jangan pakai `http://localhost` di production.

3.4 Build Production

```
cd /var/www/sayone-portal

# Build aplikasi
bun run build

# Verifikasi folder .next/ terbentuk
ls -la .next/
echo "Build completed!"
```

✅ BUILD SUKSES

Jika build berhasil, folder `.next/` akan berisi file produksi siap deploy.

4. Start Aplikasi dengan PM2

4.1 Buat Ecosystem File

```
cat > /var/www/sayone-portal/ecosystem.config.cjs << 'EOF'
module.exports = {
  apps: [{
    name: 'sayone-portal',
    script: 'node_modules/.bin/next',
    args: 'start -p 3000',
    cwd: '/var/www/sayone-portal',
    env: {
      NODE_ENV: 'production',
      PORT: 3000
    },
    instances: 1,
    autorestart: true,
    max_memory_restart: '500M',
    error_file: '/var/log/sayone-portal/error.log',
    out_file: '/var/log/sayone-portal/out.log',
    log_date_format: 'YYYY-MM-DD HH:mm:ss'
  }]
};
EOF

# Buat folder log
sudo mkdir -p /var/log/sayone-portal
sudo chown $USER:$USER /var/log/sayone-portal
```

4.2 Start Aplikasi

```
cd /var/www/sayone-portal

# Start dengan PM2
pm2 start ecosystem.config.cjs

# Cek status
pm2 status

# Lihat log
pm2 logs sayone-portal --lines 20
```

4.3 Set PM2 Auto-Start on Boot

```
# Generate startup script
pm2 startup systemd

# Ikuti instruksi yang muncul (copy-paste command yang ditampilkan)
# Contoh output:
```

```
# sudo env PATH=$PATH:/usr/bin pm2 startup systemd ...
```

```
# Save process list  
pm2 save
```

```
# Verifikasi  
pm2 list
```

✓ **APLIKASI RUNNING**

Portal sekarang berjalan di `http://localhost:3000` . Test: `curl http://localhost:3000`

5. Konfigurasi Nginx (Reverse Proxy)

5.1 Install Nginx

```
sudo apt install -y nginx
```

```
# Verifikasi  
nginx -v
```

5.2 Buat Site Configuration

```
sudo cat > /etc/nginx/sites-available/sayone-portal << 'EOF'  
server {  
    listen 80;  
    server_name portal.sayone.my.id; # Ganti dengan domain Anda  
  
    # Redirect ke HTTPS (aktifkan setelah SSL di-setup)  
    # return 301 https://$server_name$request_uri;  
  
    # Next.js app  
    location / {  
        proxy_pass http://127.0.0.1:3000;  
        proxy_http_version 1.1;  
        proxy_set_header Upgrade $http_upgrade;  
        proxy_set_header Connection 'upgrade';  
        proxy_set_header Host $host;  
        proxy_set_header X-Real-IP $remote_addr;  
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;  
        proxy_set_header X-Forwarded-Proto $scheme;  
        proxy_cache_bypass $http_upgrade;  
    }  
  
    # Static files caching  
    location /_next/static/ {  
        proxy_pass http://127.0.0.1:3000;  
        proxy_cache_valid 200 1y;  
        add_header Cache-Control "public, immutable";  
    }  
  
    # Gzip compression  
    gzip on;  
    gzip_types text/plain text/css application/json application/javascript  
text/xml;  
    gzip_min_length 1000;  
}  
EOF
```

5.3 Enable Site

```
# Enable site
sudo ln -s /etc/nginx/sites-available/sayone-portal /etc/nginx/sites-enabled/

# Test konfigurasi
sudo nginx -t

# Restart Nginx
sudo systemctl restart nginx
sudo systemctl enable nginx

# Verifikasi
sudo systemctl status nginx
```

GANTI DOMAIN

Ganti `portal.sayone.my.id` dengan domain/subdomain Anda yang sebenarnya.

6. SSL Certificate (Let's Encrypt)

6.1 Install Certbot

```
sudo apt install -y certbot python3-certbot-nginx
```

6.2 Generate SSL Certificate

```
# Generate certificate untuk domain Anda
sudo certbot --nginx -d portal.sayone.my.id

# Ikuti prompt:
# 1. Masukkan email
# 2. Setuju terms of service (Y)
# 3. Subscribe newsletter (N)
# 4. Pilih redirect HTTP → HTTPS (2 = auto redirect)
```

6.3 Verifikasi Auto-Renewal

```
# Test renewal
sudo certbot renew --dry-run

# Certbot auto-renew diatur via systemd timer
sudo systemctl status certbot.timer
```

✓ SSL AKTIF

Portal sekarang accessible via `https://portal.sayone.my.id` dengan SSL.

6.4 Update Nginx untuk HTTPS

Certbot otomatis mengubah Nginx config. Verifikasi:

```
# Lihat config yang sudah diupdate
sudo cat /etc/nginx/sites-available/sayone-portal

# Pastikan ada listen 443 ssl;
# dan redirect 80 → 443
```

7. Konfigurasi Backend CORS

Backend HRIS harus mengizinkan origin portal untuk akses API. Pastikan backend sudah dikonfigurasi:

7.1 Update CORS_ALLOWED_ORIGINS

Di server backend HRIS, set environment variable:

```
# Di /etc/environment atau application.properties backend
CORS_ALLOWED_ORIGINS=https://portal.sayone.my.id,https://hris.sayone.my.id
```

7.2 Atau Update application-prod.properties

```
# File: SayOne-HRIS/src/main/resources/application-prod.properties
spring.cors.allowed-origins=https://portal.sayone.my.id,https://hris.sayone.my.id
```

PENTING

Jika CORS tidak di-set, browser akan block request dari portal ke backend. Error: `CORS policy: No 'Access-Control-Allow-Origin' header`

7.3 Generate Portal Slug (di Admin HRIS)

1 Login Admin HRIS

Buka `https://hris.sayone.my.id/login` → login dengan akun admin

2 Buka Menu Career Portal

Navigate ke **Recruitment** → **Career Portal**

3 Generate Access Link

Di section "**Portal Access Link**", klik tombol "**Generate Access Link**"

4 Copy URL

URL akan terbentuk: `https://portal.sayone.my.id/?p=a8k2m9x7`

URL ini yang dibagikan ke candidate

5 Set Branding (Optional)

Isi Company Display Name, Logo URL, Tagline, Primary Color → Save

8. Verifikasi & Troubleshooting

8.1 Verifikasi Deployment

```
# 1. Cek PM2 running
pm2 status
# Output: sayone-portal → online ✓

# 2. Cek Nginx
sudo systemctl status nginx
# Output: active (running) ✓

# 3. Test portal lokal
curl -I http://localhost:3000
# Output: HTTP/1.1 200 ✓

# 4. Test portal via domain
curl -I https://portal.sayone.my.id
# Output: HTTP/2 200 ✓

# 5. Test API connection (dari portal ke backend)
curl https://hris.sayone.my.id/api/public/careers/SAYONE/postings
# Output: {"success":true,...} ✓
```

8.2 Troubleshooting

Masalah: Portal blank / white screen

```
# Cek PM2 logs
pm2 logs sayone-portal --lines 50

# Restart aplikasi
pm2 restart sayone-portal
```

Masalah: CORS error di browser

```
# Pastikan backend CORS_ALLOWED_ORIGINS include portal domain
# Di backend HRIS server:
echo $CORS_ALLOWED_ORIGINS
# Harus: https://portal.sayone.my.id
```

Masalah: 502 Bad Gateway

```
# Aplikasi Next.js mungkin crash
pm2 restart sayone-portal

# Cek port 3000
```

```
sudo ss -tlnp | grep 3000  
# Harus ada process next-server
```

Masalah: SSL certificate error

```
# Renew certificate manual  
sudo certbot renew  
  
# Restart Nginx  
sudo systemctl restart nginx
```

Masalah: Portal slug tidak resolve

```
# Test slug resolve endpoint  
curl https://hris.sayone.my.id/api/public/portal/YOUR_SLUG  
# Harus return: {"success":true,"data":{"companyCode":"SAYONE",...}}  
  
# Jika 404: backend belum dideploy ulang dengan CareerPortalController  
# Jika 400: slug salah atau sudah di-regenerate
```

9. Maintenance & Update

9.1 Update Portal (Pull Latest Code)

```
cd /var/www/sayone-portal

# Stop aplikasi
pm2 stop sayone-portal

# Pull latest code
git pull origin main

# Install dependencies (jika ada perubahan)
bun install

# Rebuild
bun run build

# Restart aplikasi
pm2 start sayone-portal

# Verifikasi
pm2 logs sayone-portal --lines 10
```

9.2 Backup Konfigurasi

```
# Backup env + ecosystem
cp /var/www/sayone-portal/.env.local ~/backup-portal-env-$(date +%Y%m%d)
cp /var/www/sayone-portal/ecosystem.config.cjs ~/backup-portal-ecosystem-$(date +%Y%m%d)

# Backup Nginx config
sudo cp /etc/nginx/sites-available/sayone-portal ~/backup-portal-nginx-$(date +%Y%m%d)
```

9.3 Monitoring

```
# PM2 monitoring dashboard
pm2 monit

# Real-time logs
pm2 logs sayone-portal

# Nginx access log
sudo tail -f /var/log/nginx/access.log

# Nginx error log
sudo tail -f /var/log/nginx/error.log
```


9.4 Useful Commands

Command	Fungsi
<code>pm2 status</code>	Lihat status semua aplikasi
<code>pm2 restart sayone-portal</code>	Restart portal
<code>pm2 stop sayone-portal</code>	Stop portal
<code>pm2 logs sayone-portal</code>	Lihat log real-time
<code>sudo systemctl restart nginx</code>	Restart Nginx
<code>sudo nginx -t</code>	Test Nginx config

SELESAI!

Portal kandidat SayOne HRIS sudah live di production! Candidate dapat mengakses via URL dengan slug yang dibagikan HR.